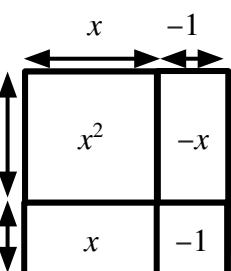


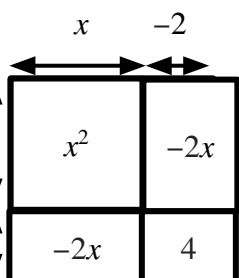


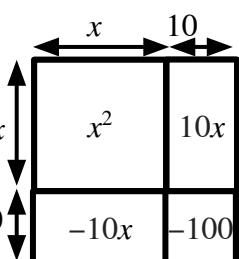
Difference of two squares

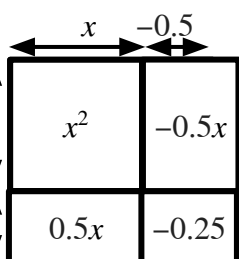
Task A: Multiplying binomials

1) Fill in the area models for these products of two binomials. Write your answer as a simplified expression

a)  $(x+1)(x-1)$
 $\equiv x^2 - x + x - 1$
 $\equiv x^2 - 1$

c)  $(x-2)(x-2)$
 $\equiv x^2 - 2x - 2x - 4$
 $\equiv x^2 - 4x - 4$

b)  $(x-10)(x+10)$
 $\equiv x^2 - 10x + 10x - 100$
 $\equiv x^2 - 100$

d)  $(x+0.5)(x-0.5)$
 $\equiv x^2 - 0.5x + 0.5x - 0.25$
 $\equiv x^2 - 0.25$

2) Which of these products of two binomials can be simplified to an expression with exactly two terms?

a) $(x+9)(x-8)$

e) $(x+7)(x+7)$

i) $(x+\frac{11}{2})(x-5.5)$

b) $(x-9)(x-9)$

f) $(x+11)(x-11)$

j) $(x-0.8)(x+\frac{4}{5})$

c) $(x-6)(x+5)$

g) $(x-2.4)(x+2.4)$

k) $(x+\frac{3}{8})(x-0.37)$

d) $(x-13)(x+13)$

h) $(x-5)(x+5.5)$

l) $(x-\frac{17}{20})(x-1.85)$

3) Match a binomial in the left column with a binomial in the middle column and find its product in the end column.

a)	$(x - 6)$	p)	$(x + 6)$	x)	$x^2 - 36$
b)	$(x - 5)$	k)	$(x - 5)$	u)	$x^2 - 10x + 25$
c)	$(x - 2)$	o)	$(x + 1)$	v)	$x^2 - x - 2$
d)	$(x + 1)$	n)	$(x - 1)$	t)	$x^2 - 1$
e)	$(x + 2)$	m)	$(x - 2)$	s)	$x^2 - 4$
f)	$(x + 4)$	i)	$(x - 9)$	w)	$x^2 - 5x - 36$
g)	$(x + 7)$	j)	$(x - 7)$	q)	$x^2 - 49$
h)	$(x + 9)$	l)	$(x - 4)$	r)	$x^2 + 5x - 36$

Task B: Identifying difference of two squares

1) Which of these products of two binomials can be written as the difference of two squares?

a) $(x + 1)(x - 4)$

e) $(x - y)(x + y)$

i) $(xy - 3)(xy + 3)$

b) $(y + 1)(x - 1)$

f) $(3a - b)(3a - b)$

j) $(3x - 2y)(-3x + 2y)$

c) $(a - 46)(a + 46)$

g) $(10 + x)(10 - x)$

$$\equiv 10^2 - x^2 \equiv 100 - x^2$$

One of the terms in each bracket needs to be the same and the other two should be a zero pair. Both of these are zero pairs.



2) Fill in the area models for these products of two binomials. Write your answer as the difference of two squares.

a)

y^2	$20y$
$-20y$	-400

$$(y + 20)(y - 20)$$

$$\equiv y^2 - 20^2$$

$$\equiv y^2 - 400$$

c)

x^2	$2xy$
$-2xy$	$-4y^2$

$$(x - 2y)(x + 2y)$$

$$\equiv x^2 - (2y)^2$$

$$\equiv x^2 - 4y^2$$

b)

x^2	$10x$
$-10x$	-100

$$(x - 10)(x + 10)$$

$$\equiv x^2 - 10^2$$

$$\equiv x^2 - 100$$

d)

a^2	$3ab$
$-3ab$	$-9b^2$

$$(a - 3b)(a + 3b)$$

$$\equiv a^2 - (3b)^2$$

$$\equiv a^2 - 9b^2$$

e)

9	$-3x$
$3x$	$-x^2$

$$(3 + x)(3 - x)$$

$$\equiv 3^2 - x^2$$

$$\equiv 9 - x^2$$

g)

x^2	$4x$
$-4x$	-16

$$(x - 4)(x + 4)$$

$$\equiv x^2 - 4^2$$

$$\equiv x^2 - 16$$

f)

$4y^2$	$6xy$
$-6xy$	$-9x^2$

$$(2y - 3x)(2y + 3x)$$

$$\equiv (2y)^2 - (3x)^2$$

$$\equiv 4y^2 - 9x^2$$

h)

$25a^2$	$15ab$
$-15ab$	$-9b^2$

$$(5a + 3b)(5a - 3b)$$

$$\equiv (5a)^2 - (3b)^2$$

$$\equiv 25a^2 - 9b^2$$

